

Day 1: Networking Devices

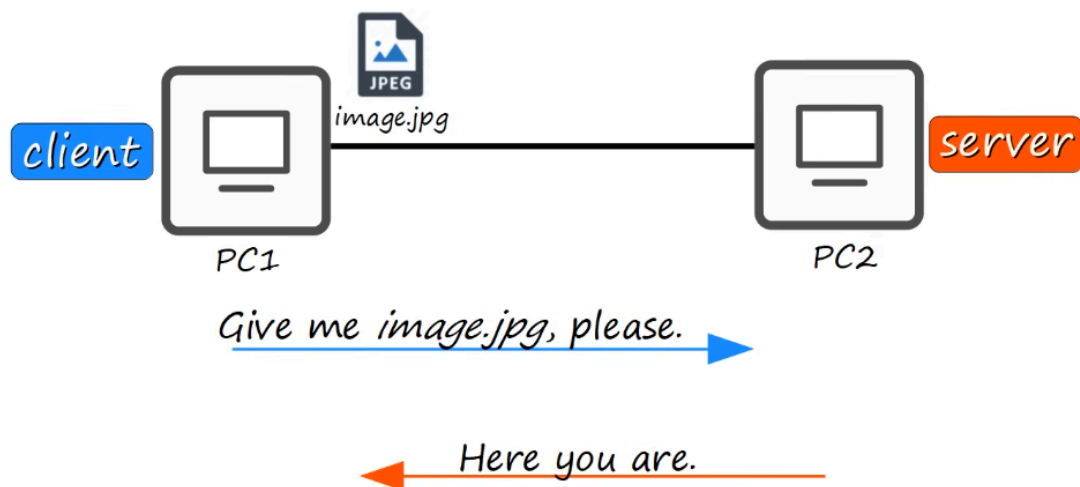
What is a network?

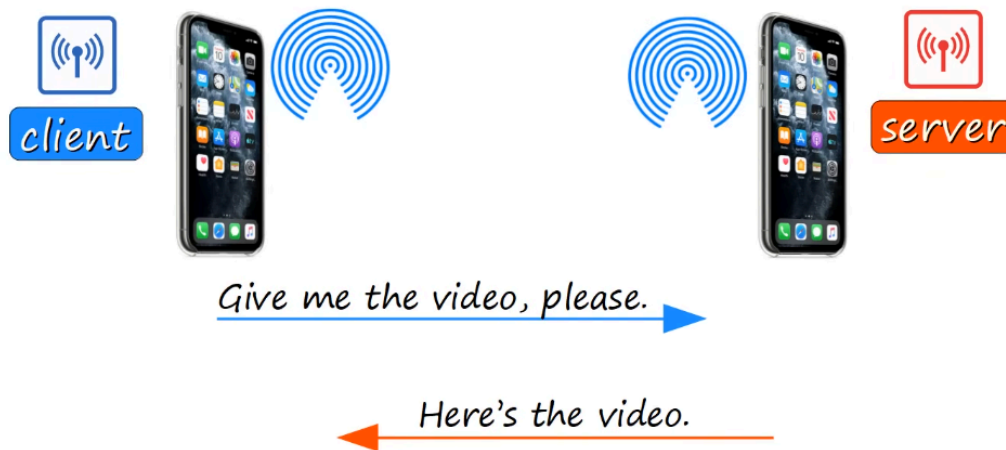
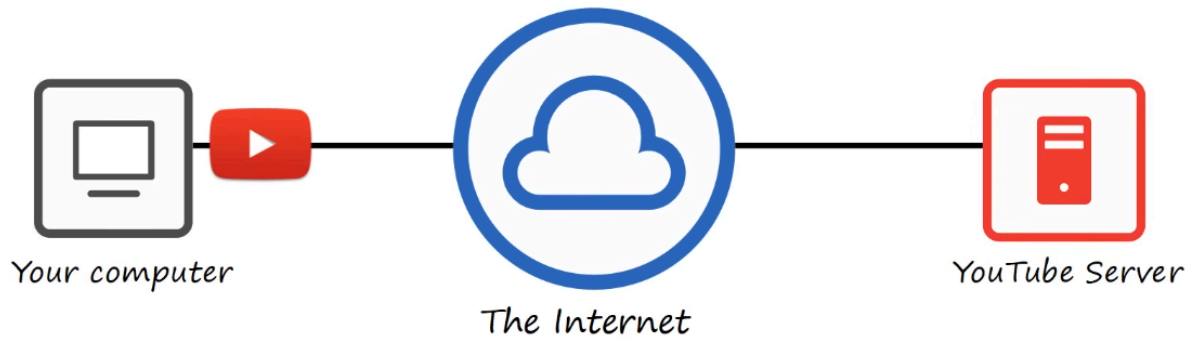
A computer network is a digital telecommunications network allows NODES to share RESOURCES.

A CLIENT is a device that accesses a service made available by a SERVER.

A SERVER is a device that provides functions or services for CLIENTS.

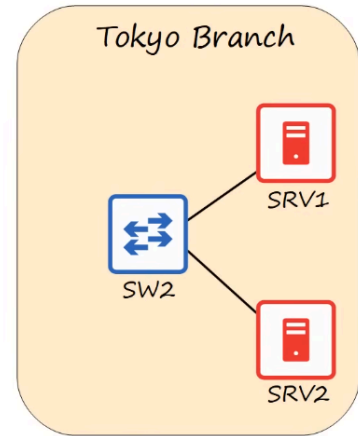
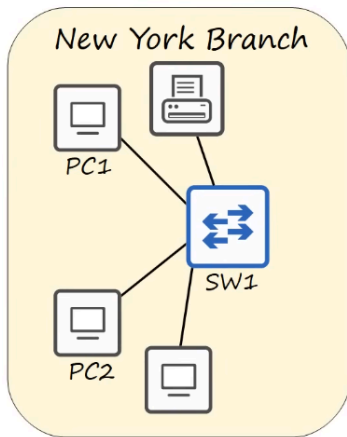
- Note : The same device can be a CLIENT in some situations and a SERVER in other situations. Ex: A Peer-to-Peer network.



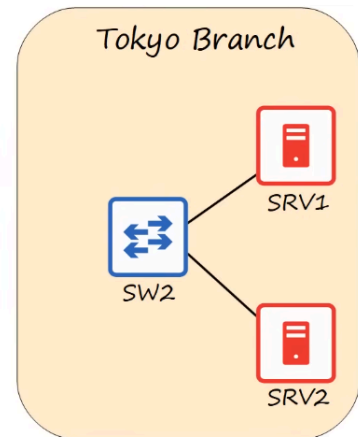
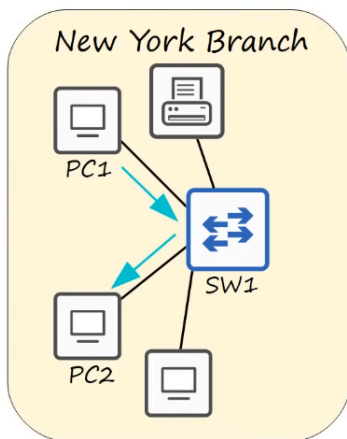


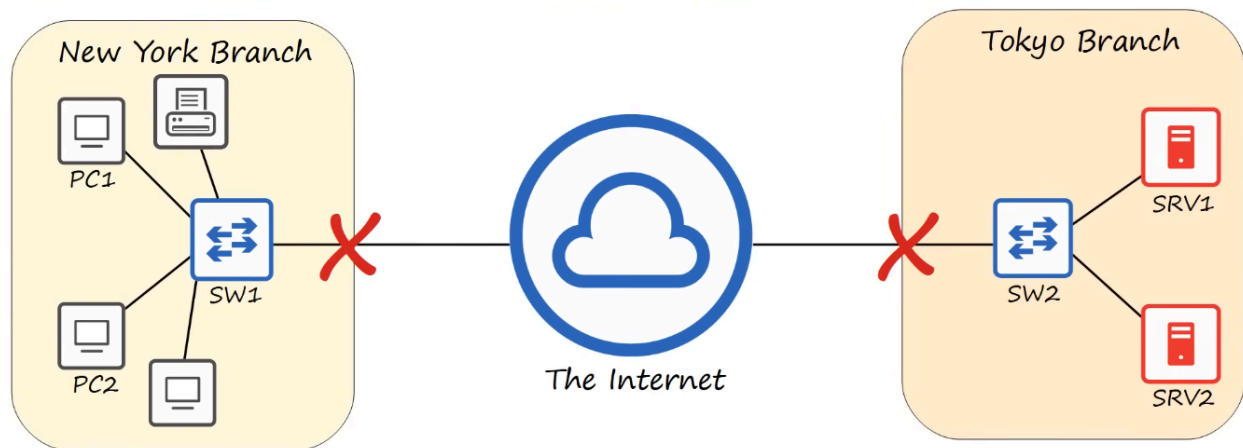
SWITCHES (Level 2):

- provide connectivity to hosts within the same LAN (Local Area Network)
- Have many network interfaces/ports for End Hosts to connect to (usually 24+).
- DO NOT provide connectivity between LANs/over the Internet.



LAN (Local Area Network)





Catalyst 9200



Catalyst 3650

ROUTERS (Level 3):

- have fewer network interfaces than switches.
- are used to provide connectivity BETWEEN LANs.
- are used to send data over the Internet.



ISR 1000



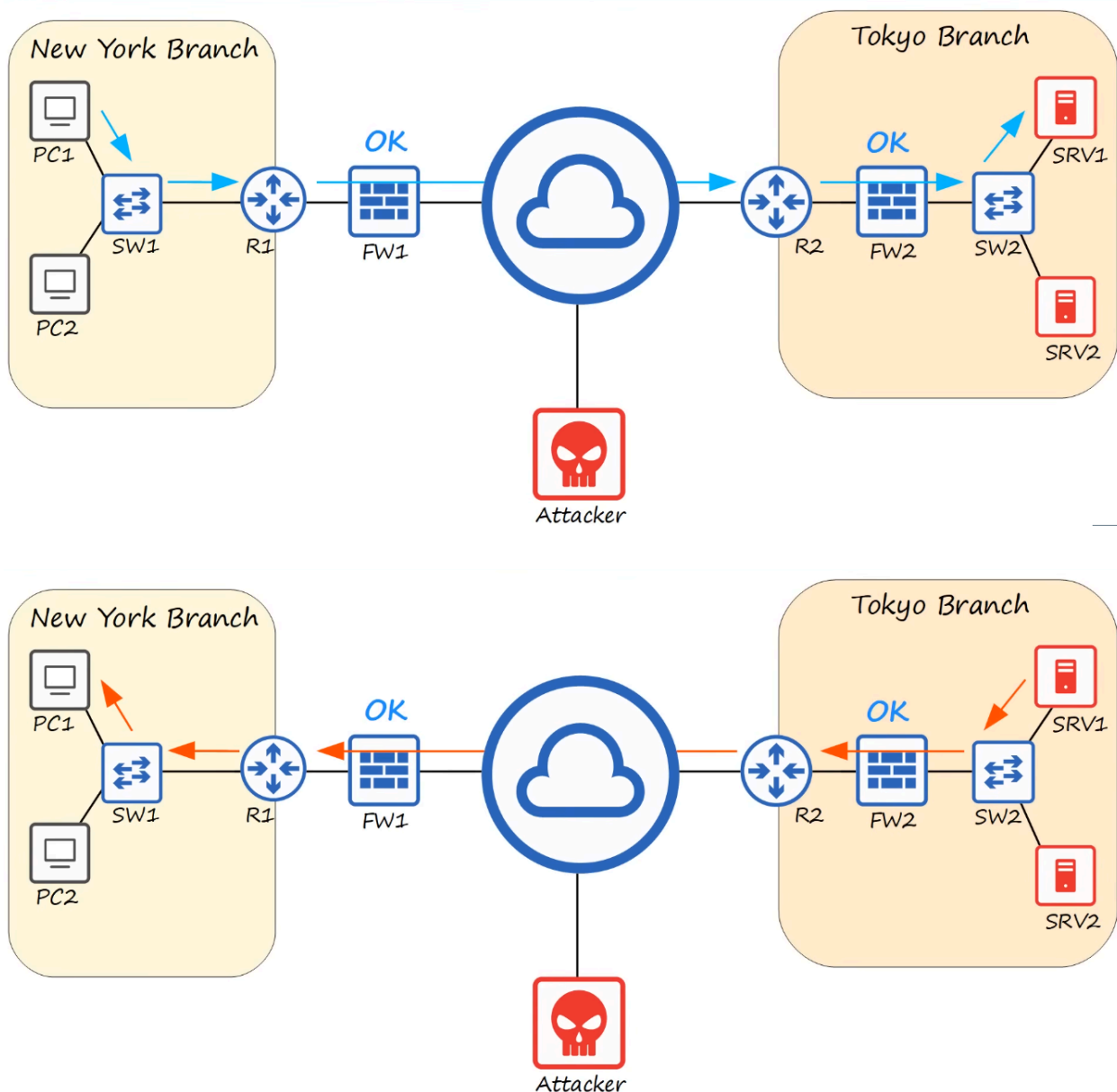
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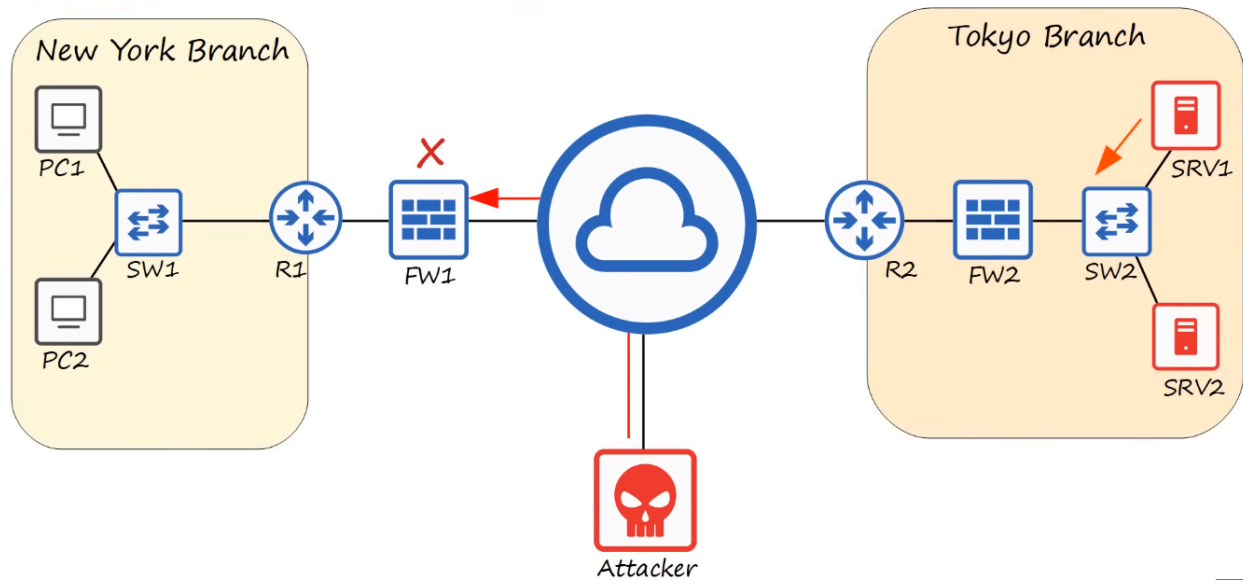


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FIREWALL (Can be Level 3,4, and 7):

- Firewalls are specialty hardware network security devices that control network traffic entering/exiting your network.
- Can be placed "inside" (FW2) or "outside" (FW1) the network.
- Monitor and control network traffic based on configured rules.
- Are known as "Next-Generation Firewalls" when they include more modern and advanced filtering capabilities.
- Host-based firewalls are software applications that filter traffic entering and exiting a host machine, like a PC.





ASA5500-X



Firepower 2100